

PART-1 : MATHEMATICS

SECTION-I (i)

1) The value of

$$\lim_{x \rightarrow \infty} \frac{2(x)^{1/2} + 3(x)^{1/3} + 4(x)^{1/4} + \dots + n(x)^{1/n}}{(2x-3)^{1/2} + (2x-3)^{1/3} + \dots + (2x-3)^{1/n}} \text{ is :-}$$

- (A) 0
- (B) 2
- (C) $\sqrt{2}$
- (D) $\frac{1}{\sqrt{3}}$

2) Let $f(x)$ be a real valued function such that $f(x) = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n^2}\right) \left(1 + \frac{2x}{n^2}\right) \dots \left(1 + \frac{nx}{n^2}\right) \forall x \in \mathbb{R}^+$ and $n \in \mathbb{N}$, then which one is CORRECT ?

- (A) $f(1) > f(1)$
- (B) $f(x) < e^x \forall x \in (0, \infty)$
- (C) $f(c) < \frac{1}{2}$ for some $c, c \in (0, \infty)$
- (D) $f(x) < x^2 \forall x \in (0, \infty)$

3) Consider the functions $f(x) = |x|^5$; $g(x) = \{\cos x\}$; $h(x) = [|\sin x|]$ (where $[.]$ denote the greatest integer function and $\{.\}$ denote fractional part function). Then which of these functions is differentiable at $x = 0$?

- (A) $f(x)$ and $g(x)$
- (B) $g(x)$ and $h(x)$
- (C) $f(x)$, $g(x)$ and $h(x)$
- (D) $f(x)$ and $h(x)$

4) The derivative of the function, $f(x) = \cos^{-1} \left(\frac{1}{\sqrt{13}} (2 \cos x - 3 \sin x) \right) + \sin^{-1} \left(\frac{1}{\sqrt{13}} (2 \cos x + 3 \sin x) \right)$ w.r.t. $\sqrt{1+x^2}$ at $x = \frac{3}{4}$ is :

- (A) $\frac{3}{2}$
- (B) $\frac{5}{2}$

(C) $\frac{10}{3}$

(D) 0

SECTION-I (ii)

1) If $f(x) = x + 3x^3 + 5x^5$ and $g = f^{-1}$, then

(A) $g'(9) = \frac{1}{35}$

(B) $g'(9) = \frac{2}{35}$

(C) $g''(9) = \frac{-118}{(35)^3}$

(D) $g''(9) = -\frac{59}{(35)^3}$

2) $f(x) = \begin{cases} 0 & ; x \text{ is irrational} \\ \frac{2}{2q^2 - q^2 + q + 5 \sin^2 q + 5} & \text{if } x = \frac{p}{q} \text{ (rational) (where HCF(p, q) = 1; } p, q > 0 \text{ and } p, q \in \mathbb{Z}) \end{cases}$

Let $f(x)$ is defined $\forall x \geq 0$ then which of the following is/are incorrect?
(where $\mathbb{Z}$ is set of integer)

(A) $f(x)$ is continuous at each irrational in $(0, \infty)$

(B) $f(x)$ is continuous at each rational in $(0, \infty)$

(C) $f(x)$ is discontinuous at each rational in $(0, \infty)$

(D) $f(x)$ is discontinuous for all x in $(0, \infty)$

3) Which of the following functions are twice differentiable at $x = 0$?

(where $[.]$ denotes greatest integer function and $\{.\}$ fractional part of function)

(A) $f(x) = x |x|$

(B) $g(x) = [x^2] \tan^{-1} x - \{x^2\} \cot^{-1} x - [x^2] \frac{\pi}{2}$

(C) $h(x) = |\sin^2 x|$

(D) $k(x) = \begin{cases} x^4 \cos\left(\frac{1}{x}\right) & ; x \neq 0 \\ 0 & ; x = 0 \end{cases}$

4) $f(x) = \cos \pi (|x| + [x])$, then

(where $[.]$ denotes greatest integer function)

(A) f is continuous at $x = 1/2$

(B) f is continuous at $x = 0$

(C) f is differentiable in $(-1, 0)$

(D) f is differentiable in $(0, 1)$

SECTION-II

1) Let $x > 0$ then $\lim_{x \rightarrow 0} \left(\sqrt{\tan x} \right)^{\sqrt{x}} + (\sec x)^{\frac{1}{x}}$ is equal to

2) For the function $f(x) = \lim_{n \rightarrow \infty} \frac{\ln \left(\left[1 + 9x^2 \right] \right) - 9^n x^{2n} \cos \left(9x^2 \right)}{1 + 9^n x^{2n}}$ number of points of discontinuity in $(-3, 3)$ is (where $[.]$ denotes greatest integer function) :-

3) If $\lim_{n \rightarrow \infty} \sin \left(\frac{1}{n^2} \right) + \sin \left(\frac{2}{n^2} \right) + \dots + \sin \left(\frac{n}{n^2} \right)$ is t , then $2t$ is equal to

4) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as

$f(x) = \begin{cases} 1 - |x|, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$ and $g(x) = f(x - 1) + f(x + 1) \forall x \in \mathbb{R}$. Find number of points of non-differentiability of $g(x)$ on $\mathbb{R}$.

5) Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$ and $h : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable functions such that $f(x) = x^3 + 3x + 2$, $g(f(x)) = x$ and $h(g(g(x))) = x$ for all $x \in \mathbb{R}$. Then the value of $2h'(2) \cdot g'(6) - h(1) \cdot h(g(2))$ is equal to

6) If $y = e^{-x} \cos x$ and $y_n + k_n y = 0$, where $y_n = \frac{d^n y}{dx^n}$ and k_n is constant $\forall n \in \mathbb{N}$, then the value of k_4 is

7) Let $f(x)$ be defined in the interval $(-\infty, \infty)$ such that $f(x) = \begin{cases} \frac{a(1-x \sin x) + b \cos x + 5}{x^2}, & x < 0 \\ 3, & x = 0 \\ \left(1 + \left(\frac{cx + dx^3}{x^2} \right) \right)^{\frac{1}{x}}, & x > 0 \end{cases}$ and $g(x)$

be defined in the interval $(c, -b)$ such that $g(x) = \begin{cases} \frac{x^2 + a}{x^2 + 1}, & c < x \leq 2 + c \\ \frac{1}{4} (x^3 - x^2 + c), & 2 + c < x \leq -b - 1 \\ \frac{9}{4} (|x + b| + |x + 2a|), & -b - 1 < x < -b \end{cases}$. If $f(x)$ is continuous at $x = 0$, then the value of $\frac{(c - a - b)d}{\ln(243)}$ is

8) Let $f(x)$ be defined in the interval $(-\infty, \infty)$ such that $f(x) = \begin{cases} \frac{a(1-x \sin x) + b \cos x + 5}{x^2}, & x < 0 \\ 3, & x = 0 \\ \left(1 + \left(\frac{cx + dx^3}{x^2} \right) \right)^{\frac{1}{x}}, & x > 0 \end{cases}$ and $g(x)$

$$g(x) = \begin{cases} \frac{x^2+a}{x^2+1} & , \quad c < x \leq 2+c \\ \frac{1}{4}(x^3-x^2+c) & , \quad 2+c < x \leq -b-1 \\ \frac{9}{4}(|x+b|+|x+2a|) & , \quad -b-1 < x < -b \end{cases}$$

be defined in the interval $(c, -b)$ such that $g(x)$ is continuous. The number of points of non-differentiability of $g(x)$ in $x \in (c, -b)$ is

PART-2 : PHYSICS

SECTION-I (i)

1) In a Young's double-slit experiment, the slit separation is $d = 0.80 \text{ mm}$ and the screen is 2.0 m from the slits. The entire arrangement is immersed in a liquid of refractive index $\mu_L = \frac{4}{3}$. The vacuum wavelength of the monochromatic light is 600 nm . When the slits are observed separately, their intensities near the central region are $I_1 = 16I_0$, $I_2 = 9I_0$. At the slits, the wave emerging from S_2 leads the wave from S_1 by a phase $\frac{\pi}{3}$. A transparent slab of refractive index $\mu_S = \frac{5}{3}$ and thickness $1.35 \mu\text{m}$ is placed normally in front of S_1 , replacing an equal thickness of the liquid. Assume negligible reflection and absorption. Find the resultant intensity at the geometrical centre O of the screen after inserting the slab.

- (A) $(25 + 12\sqrt{3})I_0$
- (B) $(25 - 12\sqrt{3})I_0$
- (C) $25I_0$
- (D) $49I_0$

2) A photographer uses a linear polarizing filter to suppress glare from a glass aquarium filled with water. The refractive indices of water and glass are $\frac{6}{5}$ and $\frac{8}{5}$, respectively. A narrow beam of unpolarized light is incident from air on the horizontal top surface of the aquarium. The photographer observes reflected light coming from two interfaces the air-water surface and the water-glass interface at the bottom. Assume the top and bottom surfaces are parallel and neglect multiple reflections. For reflection from the water-glass interface to be completely plane polarized, the angle of incidence i of the original beam in air must satisfy Brewster's condition at the water-glass interface. The value of i is closest to :

- (A) 53°
- (B) 60°
- (C) 74°
- (D) 37°

3) A high-voltage pulse generator uses a circular parallel-plate capacitor as a non-contact current monitor. Each plate has radius R and separation d . To increase the capacitance, the region $0 \leq r < \frac{R}{2}$ between the plates is completely filled with a dielectric of relative permittivity 4, while the

remaining region $\frac{R}{2} < r < R$ contains air. A voltage $V(t) = V^0 \sin \omega t$ is applied across the plates. Fringing effects are neglected. A small magnetic-field probe is placed midway between the plates at a radial distance $r = \frac{3R}{4}$ from the common axis. At an instant when $\cos \omega t = \frac{1}{2}$, the magnetic field measured by the probe is :

- (A) $\frac{\mu_0 \epsilon_0 \omega V_0 R}{8d}$
 (B) $\frac{7\mu_0 \epsilon_0 \omega V_0 R}{16d}$
 (C) $\frac{7\mu_0 \epsilon_0 \omega V_0 R}{24d}$
 (D) $\frac{3\mu_0 \epsilon_0 \omega V_0 R}{8d}$

4) A stretched string of length L , fixed at both ends, vibrates in its fifth harmonic:

$y(x, t) = A \sin\left(\frac{5\pi x}{L}\right) \cos \omega t$. The string is uniform, has linear mass density μ and is under constant tension T . At an instant when $\omega t = \frac{\pi}{6}$, a high-speed energy sensor measures the instantaneous mechanical energy stored only in the portion $\frac{L}{20} \leq x \leq \frac{L}{8}$. The sensor includes both kinetic energy and elastic potential energy of this part of the string. Neglect damping. The ratio of the energy measured by the sensor to the total mechanical energy of the entire string is :

- (A) $\frac{3\pi - 2 - \sqrt{2}}{40\pi}$
 (B) $\frac{3\pi + 2 + \sqrt{2}}{40\pi}$
 (C) $\frac{3\pi - 1 - \sqrt{2}}{40\pi}$
 (D) $\frac{3\pi - 2 + \sqrt{2}}{40\pi}$

SECTION-I (ii)

1) A mine elevator carries a package using a tapered flexible cable of length L . Its linear mass density varies as $\mu(x) = \mu_0 \left(1 + \frac{x}{L}\right)$, where x is measured upward from the package. The package has mass $M = \frac{\mu_0 L}{2}$. The elevator accelerates vertically upward with constant acceleration a . A small transverse pulse is generated at the package and travels upward along the cable. Assume the cable remains vertical and the transverse motion does not alter the tension distribution. Which statements is/are correct?

- (A) $v(x) = \sqrt{\frac{(g+a)(L+x)}{2}}$

(B) $\frac{v(L)}{v(0)} = 2$

(C) $t = 2(\sqrt{2} - 1)\sqrt{\frac{2L}{g+a}}$

(D) $t = 2 - \sqrt{2}\sqrt{\frac{L}{g}}$
If $a = g$,

2) A uniform slender elastic rod of length L , cross-sectional area S , density ρ , and Young's modulus Y is rigidly clamped at $x = 0$ and free at $x = L$. In an ultrasonic resonance experiment, its longitudinal displacement is $\xi(x, t) = \xi_0 \sin\left(\frac{9\pi x}{2L}\right) \cos \omega t$. Which of the following statements is/are correct?

(A) The rod is vibrating in its fifth allowed longitudinal mode, with frequency $f = \frac{9}{4L} \sqrt{\frac{Y}{\rho}}$.

(B) The corresponding longitudinal stress field is $\sigma(x, t) = \frac{9\pi Y \xi_0}{2L} \cos\left(\frac{9\pi x}{2L}\right) \cos \omega t$ and every displacement node is a stress antinode.

(C) At $x = \frac{L}{18}$, $\omega t = \frac{\pi}{3}$, the ratio of instantaneous kinetic-energy density to elastic potential-energy density is $\frac{u_K}{u_P} = 3$.

(D) The total instantaneous mechanical energy per unit length is independent of position x at every instant.

3) Which of the following pairs of electric and magnetic fields represent physically possible electromagnetic waves in vacuum?

(A) $\vec{E} = E_0 \cos(kx - \omega t)\hat{y}$, $\vec{B} = \frac{E_0}{c} \cos(kx - \omega t)\hat{z}$

(B) $\vec{E} = E_0 \sin(kx - \omega t)\hat{y}$, $\vec{B} = \frac{E_0}{c} \cos(kx - \omega t)\hat{z}$

(C) $\vec{E} = 2E_0 \cos kx \cos \omega t \hat{y}$, $\vec{B} = \frac{2E_0}{c} \sin kx \sin \omega t \hat{z}$

(D) $\vec{E} = 2E_0 \sin kx \cos \omega t \hat{y}$, $\vec{B} = -\frac{2E_0}{c} \cos kx \sin \omega t \hat{z}$

4) A lossless parallel-plate transmission line consists of two perfectly conducting plates of width $2w$, separated by d , with $w \gg d$. Half the width is filled with dielectric $K_1 = 2$ and the other half with dielectric $K_2 = 8$. Both are non-magnetic. The potential difference between the plates is V and equal and opposite currents I flow uniformly along the plates. Neglect edge effects. Which statements is/are correct?

(A) The electric field has the same magnitude in both dielectrics, $E_1 = E_2 = \frac{V}{d}$, but the electric energy densities satisfy $u_{E_2} = 4u_{E_1}$.

- (B) The magnetic field between the plates is $B = \frac{\mu_0 I}{2w}$, and the Poynting vector in either dielectric is $\vec{S} = \frac{VI}{2wd} \hat{z}$.
- (C) Although $\frac{K_2}{K_1} = 4$, the electromagnetic powers transported through the two dielectric-filled halves are equal: $P_1 = P_2 = \frac{VI}{2}$.
- (D) If the dielectric $K_2 = 8$ is replaced by $K_2 = 18$, keeping V and I unchanged, the electric energy stored in that half increases, but the Poynting vector there remains $\vec{S} = \frac{VI}{2wd} \hat{z}$.

SECTION-II

- 1) A laser-based optical communication receiver uses two ideal linear polarizers to control the detected signal. The incident beam has total intensity $I_0 = 320 \text{ W/m}^2$. It consists of an unpolarized component carrying 25% of the total intensity and a linearly polarized component carrying the remaining intensity. The polarization direction of the polarized component makes an angle 30° with the transmission axis of the first polarizer. A second polarizer is placed after the first, with its transmission axis at an angle θ to that of the first polarizer. The detector receives 50 W/m^2 . Find the value of $\cos^2 \theta$. Give your answer correct to two decimal places.
- 2) A narrow slit of width $6.0 \mu\text{m}$ is used in an optical wavelength-comparison system. Two monochromatic beams of wavelengths $\lambda_1 = 500 \text{ nm}$, $\lambda_2 = 750 \text{ nm}$ are incident simultaneously on the slit from the same direction. The beams make an angle i with the normal to the slit such that $\sin i = \frac{1}{2}$. A screen parallel to the slit is placed at a perpendicular distance $\sqrt{7} \text{ m}$ from it. The point where the normal through the slit center meets the screen is taken as $y = 0$. On the side of the diffraction pattern farther from $y = 0$, determine the smallest positive value of y at which a diffraction minimum of the 500 nm pattern coincides exactly with a diffraction minimum of the 750 nm pattern. Take the positive y -direction to be the direction in which the central diffraction maximum is displaced.
- 3) A convex lens of focal length 20 cm and diameter 5.0 cm is used in a Billet split-lens arrangement. A narrow monochromatic slit of wavelength 600 nm is placed 30 cm in front of the lens on its principal axis. The lens is cut into two halves along a plane containing its principal axis. A strip of width 1.0 mm is removed from the inner edge of each half, and the two remaining portions are then brought together so that their new straight edges just touch. A screen is placed 1.20 m beyond the plane containing the two coherent images of the slit. Due to the finite aperture of the two lens portions, the two emerging beams overlap only over a transverse width of 9.6 mm on the screen. The separation between the two coherent images of the slit is 6.0 mm . The central bright fringe lies at the midpoint of the overlap region, and only complete bright fringes lying entirely inside the overlap region are to be counted. Determine the total number of observable bright fringes.
- 4) In a Young's double-slit experiment, two coherent slits S_1 and S_2 are separated by $d = 0.600 \text{ mm}$ and emit monochromatic light of wavelength $\lambda = 600 \text{ nm}$. The locus of the 501^{st} bright fringe on one side of the perpendicular bisector is a hyperbola having S_1 and S_2 as its foci. Taking the midpoint of

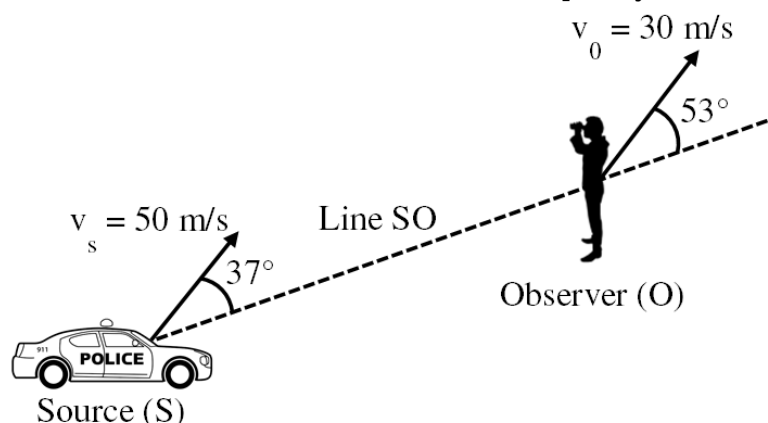
the slits as the origin, the x-axis along S_1S_2 and the y-axis along the perpendicular bisector, a point P in this bright-fringe hyperbola has $x = 0.300$ mm. Determine the positive y-coordinate of P, in mm. Do not use the usual distant-screen approximation.

5) A straight railway track of length 200 m, extending from $x = -100$ m to $x = +100$ m, carries a large number of identical mutually incoherent sound sources uniformly distributed along it. Each infinitesimal element may be treated as an isotropic point source. A sound-level meter is placed on the perpendicular bisector of the track at a distance $100\sqrt{3}$ m from its midpoint. The meter records a total sound level of 80.00 dB. When the railway sources are switched off, the background sound level is 70.00 dB. The meter is then moved to a distance 100 m from the midpoint. Assume the background intensity remains unchanged and atmospheric absorption is negligible. Given that

$$\frac{I_{\text{railway at 100m}}}{I_{\text{railway at } 100\sqrt{3}\text{m}}} = \frac{3\sqrt{3}}{2} \approx 2.60$$

, determine the total sound level at the new position. Use : $\log_{10}(2.44) \approx 0.387$.

6) A police siren of frequency 800 Hz is mounted on a vehicle moving in still air with speed 50 m/s. At a certain instant, an observer is located at point O, and the line joining the source S to the observer makes an angle 37° with the source velocity. At the same instant, the observer moves with speed 30 m/s along a direction making 53° with the line OS, with its component directed away from the source. Take the speed of sound as 330 m/s. Determine the frequency (in Hz) heard by the



observer at that instant.

7) A supersonic aircraft flies horizontally at a constant altitude of 3.0 km above a straight highway. The speed of sound in still air is 300 m/s. At the instant the aircraft passes directly above an observer, the observer starts moving along the highway in the same direction as the aircraft with a constant speed of 100 m/s. The observer hears the sonic boom exactly 10.0 s after the aircraft was directly overhead. Assume that the aircraft maintains constant speed and altitude, the air is stationary, and the shock front forms an ideal Mach cone. Determine the Mach number M of the aircraft. Give the answer correct to two decimal places.

8) In a sonometer experiment, the tension in a wire is provided by an aluminium block of density 2.70×10^3 kg m⁻³ suspended over a frictionless pulley. With the block completely in air, a tuning fork produces resonance when a length 0.90 m of the wire vibrates in 2 loops. The block is then completely immersed in an unknown liquid without touching the container. With the same tuning fork, resonance is obtained when the vibrating length is adjusted to 1.10 m, and the wire now vibrates in 3 loops. Neglect the density of air, surface-tension effects, and the mass of the connecting thread. Assume the linear mass density of the sonometer wire remains unchanged.

Determine the density of the liquid in units of 10^3 kg m^{-3} . Give your answer correct to two decimal places.

PART-3 : CHEMISTRY

SECTION-I (i)

1)

To the dissociation $2\text{N}_2\text{O}_5 \rightleftharpoons 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$, $\text{O}_2(\text{g})$ was added, maintaining its partial pressure constant. Which of the following statement is correct?

- (A) K_p increases and equilibrium shifts backwards.
- (B) K_p remains constant and the equilibrium shifts forwards.
- (C) K_p remains constant and the equilibrium remains undisturbed.
- (D) K_p remains constant and the equilibrium shifts backwards.

2) A 100 mL aqueous solution of 0.1M CuSO_4 is electrolyzed using inert platinum electrodes. The process is stopped when the solution's pH reaches 1.0. Assume the initial pH was 7.0 and the volume of the solution remains constant throughout the process. (Given: Atomic mass of Cu = 63.5, 1 F = 96500 C)

Choose the incorrect statement about the above process:

- (A) The mass of copper deposited at the cathode is 0.3175 g
- (B) The total volume of gases evolved at the electrodes at 1 atm, 0°C is 168 mL
- (C) The total charge passed through the solution is 965 C
- (D) The concentration of Cu^{2+} ions remaining in the solution is 0.05 M

3) How is an exhausted synthetic cation-exchange resin containing bound Ca^{+2} and Mg^{2+} ions regenerated back to its active hydrogen form ?

- (A) Treating with a concentrated solution of NaCl
- (B) Boiling the resin bed vigorously in distilled water
- (C) Washing with a concentrated aqueous solution of NaOH
- (D) Flushing with a concentrated solution of dilute HCl or H_2SO_4

4) In the Hall-Heroult process, aluminium is produced by the electrolysis of molten Al_2O_3 . How many seconds would it take to produce enough aluminium by the Hall-Heroult process to make a case of 27 cans of aluminium, if each can uses 5.0 g of Al. A current of 9650A is employed and the current efficiency of the cell is 100% ? (Atomic wt. : Al = 27)

- (A) 300
- (B) 450
- (C) 150
- (D) 75

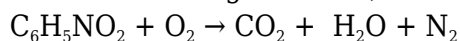
SECTION-I (ii)

1) Given pK_{a1} and pK_{a2} of H_2CO_3 are 6.35 and 10.33 respectively, choose the CORRECT Statement(s)?

[$\log 2 = 0.3$; $\log 3 = 0.48$; $\log 5 = 0.7$]

- (A) Upon mixing equal volumes of 0.1 M Na_2CO_3 , and 0.2 M H_2CO_3 , a solution of $pH < 7$ is formed.
- (B) Upon mixing equal volumes of 0.2 M Na_2CO_3 , and 0.1 M H_2CO_3 , a solution of pH more than 7 is formed
- (C) Upon mixing equal volumes of 0.1 M Na_2CO_3 , and 0.1 M H_2CO_3 , solution of pH more than 7 is formed
- (D) Upon mixing equal volumes of 0.1 M H_2CO_3 , and 0.2 M $NaOH$, pH of solution obtained is more than 10 but less than 12

2) For the following reaction ;

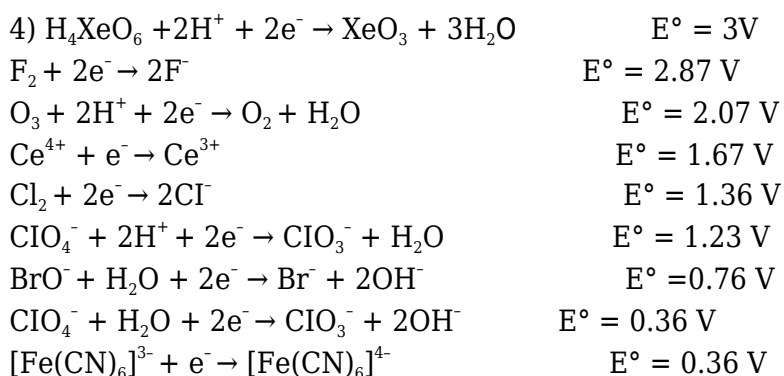


Choose the correct statement (s)

- (A) Number of electrons lost by one molecule of $C_6H_5NO_2$ is 25
- (B) One mole $C_6H_5NO_2$ required 11.2 mole oxygen atoms for complete oxidation.
- (C) One mole $C_6H_5NO_2$ on combustion give 22.4 litre $N_2(g)$ at 1 atm and 273K
- (D) One mole of $C_6H_5NO_2$ on combustion give 22.4 litre $H_2O(l)$ at 1 atm & 273K

3) Consider the cell $Cd(s) | Cd^{2+}(1.0M) || Cu^{2+}(1.0M) | Cu(s)$. If we want to make a cell with a less positive voltage using the same substances, we should

- (A) Increase both, $[Cd^{+2}]$ and $[Cu^{+2}]$, to 2.00 M
- (B) Increase only the $[Cd^{+2}]$ to 2.00 M
- (C) Decrease both, $[Cd^{+2}]$ and $[Cu^{+2}]$, to 0.100 M
- (D) Decrease only the $[Cd^{+2}]$ to 0.100 M



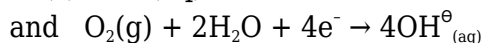
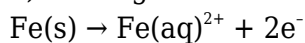
Based on the above data, which of the following statement(s) is/are correct?

- (A) F_2 is better oxidizing agent than H_4XeO_6
- (B) Ozone can oxidise Cl_2
- (C) ClO_4^- is better oxidizing agent in basic medium than in acidic medium

(D) $[\text{Fe}(\text{CN}_6)]^{4-}$ ion can be easily oxidized by Ce^{4+} and BrO^-

SECTION-II

1) Rusting of iron involves the following electrode processes:



The standard electrode potentials are : $E_{\text{Fe}^{2+}/\text{Fe}}^\circ = -0.44 \text{ V}$, $E_{\text{O}_2/\text{OH}^-}^\circ = 0.40 \text{ V}$

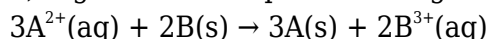
$$\frac{RT \ln 10}{F}$$

Take $\frac{RT \ln 10}{F} = 0.06$, $\log 2 = 0.31$, $\log 3 = 0.48$, $\log 5 = 0.7$

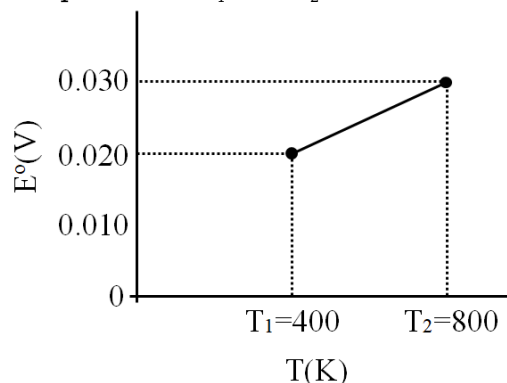
Consider temperature to be 25°C , with $K_w = 10^{-14} \text{ M}^2$. Density of water, as well as aqueous solution = 1 g mL^{-1} . Atmospheric pressure is 1 bar, and the atmosphere is 79% N_2 , 20% O_2 and 1% other gases by volume. Calculate the electrode potential (in V) for rusting, (as described above) if the water is pH neutral and $[\text{Fe}^{2+}]$ is 0.01M.

2) A 0.2 M solution of HOCl ($\text{pK}_a = 7.5$) is titrated with 0.2 M NaOH . The pH after addition of 20 mL NaOH is found to be 7.50. What will be the pH of solution after 40 mL NaOH solution has been added? (Use : $\log 3 = 0.48$ and $\log 5 = 0.7$)

3) A galvanic cell operates using the reaction:



where all species at their standard states. The standard EMF of the cell (E°) is measured at two temperatures T_1 and T_2 , and the variation is shown in the figure below:



Given : $F = 96500 \text{ C / mol}$

Find the magnitude of ΔH° in kJ/mol at 400K.

4) A substance X decomposes according to $2\text{X(s)} \rightleftharpoons 2\text{Y(g)} + \text{Z(g)}$.

At 300 K, the total pressure at equilibrium is 3 atm. At 600 K, the total pressure at equilibrium is 12 atm.

The standard Gibbs free energy change at 600 K is ΔG_2° and at 300 K is ΔG_1° . If $\frac{\Delta G_1^\circ}{300} - \frac{\Delta G_2^\circ}{600} = R \ln(m)$, find the value of 'm'.

5) Solubility product constant of a sparingly soluble salt MCl_2 is 4×10^{-12} at 25°C . Also, at 25°C , solubility of MCl_2 in an aqueous solution of CaCl_2 is 4×10^8 times less compared to its solubility in pure water. Hence, concentration (molarity) of CaCl_2 solution is

6) Resistance of 0.2 M solution of an electrolyte is $50\ \Omega$. The specific conductance of the solution is $1.4\ \text{S cm}^{-1}$. The resistance of 0.5 M solution of the same electrolyte is $280\ \Omega$. The molar conductivity of 0.5 M solution of electrolyte in $\text{S cm}^2 \text{mol}^{-1}$ is _____.

7) Using the given information $\text{ClO}^-(\text{aq}) \xrightarrow{0.4\ \text{volt}} \frac{1}{2}\text{Cl}_2(\text{aq}) \xrightarrow{1.4\ \text{volt}} \text{Cl}^-(\text{aq})$

Value of $\left\{ -\frac{10\Delta G^\circ}{F} \right\}$ in volts for the reaction $\text{Cl}_2(\text{aq}) + 2\text{OH}^-(\text{aq}) \rightleftharpoons \text{Cl}^- + \text{H}_2\text{O}(\text{aq}) + \text{ClO}^-(\text{aq})$ is _____.

8) MO_2 undergoes disproportion reaction into MO_4^- and M^{x+} ions in acidic medium. If the mole ratio of MO_2 undergoes oxidation and reduction is 2 : 3 the value of 'x' is _____

ANSWER KEYS

PART-1 : MATHEMATICS

SECTION-I (i)

Q.	1	2	3	4
A.	C	B	D	C

SECTION-I (ii)

Q.	5	6	7	8
A.	A,C,D	B,D	B,C,D	A,C,D

SECTION-II

Q.	9	10	11	12	13	14	15	16
A.	2.00	2.00	1.00	5.00	79.00	4.00	1.00	2.00

PART-2 : PHYSICS

SECTION-I (i)

Q.	17	18	19	20
A.	A	C	B	A

SECTION-I (ii)

Q.	21	22	23	24
A.	A,C	A,B,C	A,C,D	A,B,C,D

SECTION-II

Q.	25	26	27	28	29	30	31	32
A.	0.22 to 0.23	3.00	79.00	0.45	83.87	860.70 to 860.72	1.66 to 1.67	0.91

PART-3 : CHEMISTRY

SECTION-I (i)

Q.	33	34	35	36
A.	B	B	D	C

SECTION-I (ii)

Q.	37	38	39	40
A.	A,B,C,D	A	B	B,D

SECTION-II

Q.	41	42	43	44	45	46	47	48
A.	1.30 to 1.31	10.25	5.79	64.00	2.00	500.00	10.00	2.00

SOLUTIONS

PART-1 : MATHEMATICS

1)

Given,

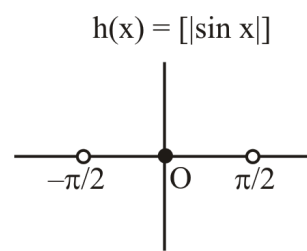
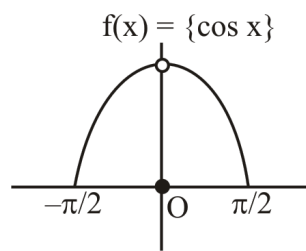
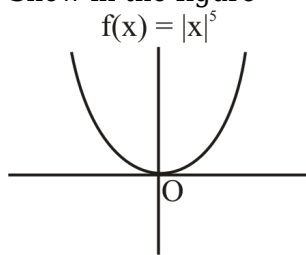
$$\begin{aligned} & \lim_{x \rightarrow \infty} \frac{2(x)^{1/2} + 3(x)^{1/3} + 4(x)^{1/4} + \dots + n(x)^{1/n}}{(2x-3)^{1/2} + (2x-3)^{1/3} + \dots + (2x-3)^{1/n}} \\ &= \lim_{h \rightarrow 0} \frac{2(h)^{-1/2} + 3(h)^{-1/3} + \dots + n(h)^{-1/n}}{h^{-1/2}(2-3h)^{1/2} + h^{-1/3}(2-3h)^{1/3} + \dots + h^{-1/n}(2-3h)^{1/n}} \left[\text{Put } x = \frac{1}{h} \right] \\ &= \lim_{h \rightarrow 0} \frac{2 + 3h^{\left(\frac{1}{2}-\frac{1}{3}\right)} + 4h^{\left(\frac{1}{2}-\frac{1}{4}\right)} + \dots + nh^{\left(\frac{1}{2}-\frac{1}{n}\right)}}{(2-3h)^{1/2} + h^{\left(\frac{1}{2}-\frac{1}{3}\right)}(2-3h)^{1/3} + \dots + h^{\left(\frac{1}{2}-\frac{1}{n}\right)}(2-3h)^{1/n}} \\ &= \frac{2+0+0+\dots+0}{2^{1/2}+0+0+\dots} = \sqrt{2} \end{aligned}$$

2) For $1 \leq k \leq n$

$$\begin{aligned} \left(1 + \frac{x}{n}\right)^{k/n} &\leq \left(1 + \frac{kx}{n^2}\right) \leq \left(1 + \frac{x}{n^2}\right)^k \\ \left(1 + \frac{x}{n}\right)^{\frac{n+1}{2}} &\leq P_n \leq \left(1 + \frac{x}{n^2}\right)^{\frac{n(n+1)}{2}} \\ e^{x/2} &\leq \lim_{n \rightarrow \infty} P_n \leq e^{x/2} = f(x) = e^{x/2} \end{aligned}$$

3)

Show in the figure



$$\begin{aligned} 4) \text{ Put } \cos \phi &= \frac{2}{\sqrt{13}}; \sin \phi = \frac{3}{\sqrt{13}}; \tan \phi = \frac{3}{2} \quad y = \cos^{-1}\{\cos(x + \phi)\} + \sin^{-1}\{\cos(x - \phi)\} \\ &= \cos^{-1}\{\cos(x + \phi) + \frac{\pi}{2}\} - \cos^{-1}\{\cos(\phi - x)\} \quad (\text{think !}) \\ &= x + \phi + \frac{\pi}{2} - \phi + x \\ y &= 2x + \frac{\pi}{2}; \quad z = \sqrt{1+x^2} \\ &\text{now compute } \frac{dy}{dz} \end{aligned}$$

$$\begin{aligned}
 5) \quad & g(9) = 1 \Rightarrow f(1) = 9 \\
 & \Rightarrow g'(9) = \frac{1}{f'(1)} = \frac{1}{35} \\
 & \Rightarrow g''(9) = -\frac{f''(1)}{(f'(1))^3} = -\frac{118}{(35)^3}
 \end{aligned}$$

6)

$$\text{Let } x = \sqrt{3}$$

$$f(\sqrt{3}) = 0$$

$$\sqrt{3} = 1.732050807 \dots$$

As the decimal part increases then in the expression $\frac{p}{q}, q$ becomes very large

$$\text{So, } \frac{2q^3 - q^2 + q + \sin^2 q + 5}{2} \rightarrow 0$$

$$\lim_{x \rightarrow \sqrt{3}} f(x) = 0$$

Hence, $x \rightarrow \sqrt{3}$

Thus, $f(x)$ is continuous at each irrational

$$7) \quad f''(x) = \begin{cases} 2; x > 0 \\ -2; x < 0 \end{cases} \Rightarrow \text{not differentiable}$$

$$g(x) = \begin{cases} -x^2 \cot^{-1} x; & x > 0 \\ -x^2 \cot^{-1} x; & x < 0 \\ 0; & x = 0 \end{cases} \Rightarrow \text{twice differentiable}$$

$$h(x) = \sin^2 x \rightarrow \text{differentiable}$$

$$k'(x) = \begin{cases} 4x^3 \cdot \cos \frac{1}{x} + x^2 \cdot \sin \frac{1}{x}; & x \neq 0 \\ 0; & x = 0 \end{cases}$$

twice differentiable

$$8) \quad f(x) = \cos \pi (|x| + [x])$$

$$f(0) = 1, f(0 - 0) = \cos \pi (-1) = -1$$

$$f(1/2) = \cos \pi \left(\frac{1}{2} \right) = 0, f\left(\frac{1}{2} - 0 \right) = \cos \pi \left(\frac{1}{2} \right) = 0$$

$$f\left(\frac{1}{2} + 0 \right) = \cos \frac{\pi}{2} = 0$$

$$\text{for } x \in (0, 1), f(x) = \cos \pi x$$

$$\text{for } x \in (-1, 0), f(x) = \cos \pi (-x - 1) = -\cos \pi x.$$

$$9) \quad \lim_{x \rightarrow 0^+} \left(\sqrt{\frac{\tan x}{x}} \right)^{\sqrt{x}} \sqrt{x}^{\sqrt{x}} = 1$$

$$10) \quad f(x) = \begin{cases} 0 & 9x^2 < 1 \\ \frac{\ln 2 - \cos 1}{2} & 9x^2 = 1 \\ -\cos 9x^2 & 9x^2 > 1 \end{cases}$$

discontinuous at $x = \pm \frac{1}{3}$

$$11) \quad \sin\left(\frac{1}{n^2}\right) + \sin\left(\frac{2}{n^2}\right) + \dots + \sin\left(\frac{n}{n^2}\right) \text{ be } S_n$$

$$\text{So, } \sum \frac{1}{n^2} - \frac{1}{6} \sum \frac{n^3}{n^6} < S_n < \frac{1}{n^2} + \frac{2}{n^2} + \dots + \frac{n}{n^2}$$

(Using $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$)

$$12) \quad f(x-1) = \begin{cases} 1 - |x-1| & ; x \in [0, 2] \\ 0 & ; x \in (-\infty, 0) \cup (2, \infty) \end{cases}$$

$$f(x+2) = \begin{cases} 1 - |x+1| & ; x \in [-2, 0] \\ 0 & ; x \in (-\infty, -2) \cup (0, \infty) \end{cases}$$

$$g(x) = f(x-1) + f(x+2) = \begin{cases} 0 & ; x \in (-\infty, -2) \\ 1 - |x+1| & ; x \in [-2, 0] \\ 0 & ; x = 0 \\ 1 - |x-1| & ; x \in (0, 2] \\ 0 & ; x \in (2, \infty) \end{cases}$$

$g(x)$ is not differentiability at $x = -2, -1, 0, 1, 2$

$$13) \quad g(f(x)) = x \square f(x) \text{ \& } g(x) \text{ are inverse function to each other}$$

$$g'(f(x)) = \frac{1}{f'(x)}$$

$$\text{put } x = 1$$

$$g'(6) = \frac{1}{6}$$

$$f(0) = 2 \square g(2) = 0$$

$$h(g(g(x))) = x$$

$$\text{Put } x = f(x)$$

$$h(g(x)) = f(x) \square h(g(2)) = f(2) = 8 + 6 + 2 = 16$$

$$h(x) = f(f(x)) \square h'(2) = f'(f(2)) \quad f'(2) = 771 \times 15$$

$$h(1) = f(f(1)) = f(6) = 236$$

$$2h'(2)g'(6) - h(1) \cdot h(g(2))$$

$$2 \times 11565 \times \frac{1}{6} - (236) \times 16 = 79$$

$$14)$$

$$y = e^{-x} \cos x$$

$$y_1 = -[e^{-x} \cos x + e^{-x} \sin x]$$

$$\Rightarrow y_1 = -\sqrt{2} e^{-x} \cos \left(x - \frac{\pi}{4} \right)$$

$$y_2 = +\sqrt{2} \left[e^{-x} \cos \left(x - \frac{\pi}{4} \right) + e^{-x} \sin \left(x - \frac{\pi}{4} \right) \right]$$

$$y_2 = 2e^{-x} \cos \left(x - \frac{\pi}{2} \right) = 2e^{-x} \sin x$$

$$y_3 = 2(e^{-x} \cos x - e^{-x} \sin x)$$

$$y_3 = 2\sqrt{2} e^{-x} \cos \left(x + \frac{\pi}{4} \right)$$

$$y_4 = -2\sqrt{2} \left[e^{-x} \cos \left(x + \frac{\pi}{4} \right) + e^{-x} \sin \left(x + \frac{\pi}{4} \right) \right]$$

$$y_4 = -4e^{-x} \cos \left(x + \frac{\pi}{4} - \frac{\pi}{4} \right)$$

$$y_4 = -4e^{-x} \cos x \Rightarrow y_4 = -4y \Rightarrow y_4 + 4y = 0$$

$$k_4 = 4 \quad y_8 + 4y_4 = 0 \Rightarrow y_8 - 16y = 0$$

$$15) \quad f(0^-) = \lim_{h \rightarrow 0} \frac{a(1 - h \sinh) + b \cosh + 5}{h^2}$$

$$= \lim_{h \rightarrow 0} \frac{a \left(1 - h \left(h - \frac{h^3}{3!} + \frac{h^5}{5!} + \dots \right) \right) + b \left(1 - \frac{h^2}{2!} + \frac{h^4}{4!} \right) + 5}{h^2}$$

$$= \lim_{h \rightarrow 0} \frac{(a + b + 5) + h^2 \left(-a - \frac{b}{2} \right) + \dots}{h^2}$$

$$\text{For existence of limit } a + b + 5 = 0 \quad \dots(1)$$

& for continuous function

$$f(0^-) = f(0) = f(0^+)$$

$$-a - \frac{b}{2} = 3 \quad \dots(2)$$

from (1) and (2)

$$\boxed{a = -1, b = -4}$$

Now

$$f(0^+) = \lim_{h \rightarrow 0} \left[1 + \frac{h(c + dh^2)}{h^2} \right]^{\frac{1}{h}}$$

for existence of limit $\boxed{c = 0}$

$$= \lim_{h \rightarrow 0} [1 + dh]^{\frac{1}{h}} = e^d \quad \lim_{x \rightarrow 0} (1 + x)^{\frac{1}{x}} = e$$

Now $e^d = 3$

$$\boxed{d = \ln 3}$$

$$16) \text{ Now } f(x) = \begin{cases} \frac{x^2-1}{x^2+1} & , \quad 1 < x \leq 2 \\ \frac{x^3-x^2}{4} & , \quad 2 < x \leq 3 \\ \frac{9}{4} (|x-4| + |x-2|) & , \quad 3 < x < 4 \end{cases}$$

$$f'(2^+) = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h}$$

$$\begin{aligned}
&= \lim_{h \rightarrow 0} \frac{\frac{(2+h)^2((2+h)-1)}{4} - \frac{3}{5}}{h} \\
&= \lim_{h \rightarrow 0} \frac{5(h^3 + 5h^2 + 8h) - \frac{3}{5}}{20h} \\
&= \text{Does not exist}
\end{aligned}$$

Hence it is not differentiable at $x = 2$.

$$\begin{aligned}
f'(3^+) &= \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} \\
&= \lim_{h \rightarrow 0} \frac{\frac{9}{4}(|h-1| + |h+1|) - \frac{18}{4}}{h} \\
&= \frac{9}{4} \lim_{h \rightarrow 0} \frac{-h+1+h+1-2}{h} = 0 \\
f'(3^-) &= \lim_{h \rightarrow 0} \frac{f(3-h) - f(3)}{-h} \\
&= \lim_{h \rightarrow 0} \frac{\frac{(3-h)^3 - (3-h)^2}{4} - \frac{18}{4}}{-h} \\
&= \frac{1}{4} \lim_{h \rightarrow 0} \frac{(3-h)^3(3-h-1) - 18}{h} \\
&= \frac{1}{4}(-21) = \frac{-21}{4} \\
f'(3^-) &\neq f'(3^+)
\end{aligned}$$

Hence $f(x)$ is not differentiable at $x = 3$

PART-2 : PHYSICS

17)

Correct Answer is (A)

18)

Correct Answer is (C)

19)

Correct Answer is (B)

20)

Correct Answer is (A)

21)

Correct Answer is (A,C)

22)

Correct Answer is (A,B,C)

23)

Correct Answer is (A,C,D)

24)

Correct Answer is (A,B,C,D)

25)

Correct Answer is (0.22 to 0.23)

26)

Correct Answer is (3.00)

27)

Correct Answer is (79.00)

28)

Correct Answer is (0.45)

29)

Correct Answer is (83.87)

30)

Correct Answer is (860.70 to 860.72)

31)

Correct Answer is (1.66 to 1.67)

32)

Correct Answer is (0.91)

PART-3 : CHEMISTRY

33) Due to addition of O_2 at constant pressure volume of container will increase, so reaction will move in forward direction.

34) Initial conditions: 100 mL of 0.1 M $CuSO_4$ (0.01 moles of Cu^{2+})

Anode reaction : $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$

pH change : Final pH = 1.0 $\Rightarrow [H^+] = 0.1$ M

Moles of H^+ produced = $0.1 \text{ M} \times 0.100 \text{ L} = 0.01 \text{ mol}$

Charge passed (Q) : Moles of $e^- = 0.01 \text{ mol} \Rightarrow Q = 0.01 \times 96500 = 965 \text{ C}$

Cathode reaction : $Cu^{2+} + 2e^- \rightarrow Cu(s)$

Moles of Cu deposited = $\frac{0.01}{2} = 0.005 \text{ mol}$

Mass of Cu = $0.005 \times 63.5 = 0.3175 \text{ g}$

Gas evolved (O_2): Moles of $O_2 = \frac{0.01}{4} = 0.0025 \text{ mol}$

Volume at STP = $0.0025 \times 22400 \text{ mL} = 56 \text{ mL}$

Remaining $[Cu^{2+}]$:

Remaining moles = $0.01 - 0.005 = 0.005 \text{ mol}$

Concentration = $\frac{0.005 \text{ mol}}{0.1 \text{ L}} = 0.05 \text{ M}$

35) Cation exchangers (R-H) are regenerated by washing with strong aqueous acid solutions, replacing bound divalent metal ions with protons.

36) Total Aluminum Needed: 27 cans $\times$ 5.0 g/can = 135 g.

Moles of Al: $\frac{135\text{g}}{27\text{g/mol}} = 5 \text{ mol}$.

Reduction Reaction: $Al^{3+} + 3e^- \rightarrow Al$

Total moles of $e^- = 5 \times 3 = 15 \text{ mol}$

Total Charge (Q): $15 \times 96500 = 1,447,500 \text{ C}$.

Time (t): $t = \frac{Q}{I} = \frac{1,447,500}{9650A} = 150 \text{ seconds}$

37) A) Equal volumes of 0.1 M Na_2CO_3 and 0.2 M H_2CO_3 :

- Forms equal moles of $NaHCO_3$ (0.1 mol) and remaining H_2CO_3 (0.1 mol).
- Acidic buffer: $pH = pK_{a1} + \log(1) = 6.35 < 7$ (A is Correct).

B) Equal volumes of 0.2 M Na_2CO_3 and 0.1 M H_2CO_3 :

- Forms 0.1 mol $NaHCO_3$ and leaves 0.1 mol Na_2CO_3 .
- Basic buffer: $pH = pK_{a2} + \log(1) = 10.33 > 7$ (B is Correct).

C) Equal volumes of 0.1 M Na_2CO_3 and 0.1 M H_2CO_3 :

- Completely forms amphoteric HCO_3^- .

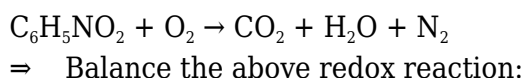
$$pH = \frac{pK_{a1} + pK_{a2}}{2} = \frac{6.35 + 10.33}{2} = 8.34 > 7 \text{ (C is Correct).}$$

D) Equal volumes of 0.1 M H_2CO_3 and 0.2 M $NaOH$:

- Forms pure 0.05 M Na_2CO_3 solution.

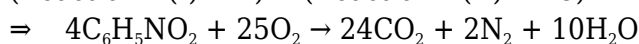
$$pH = 7 + \frac{1}{2} pK_{a2} + \frac{1}{2} \log C = 7 + 5.165 + \frac{1}{2} \log(0.05) = 12.165 - 0.65 = 11.515 \text{ (between 10 and 12) (D is Correct).}$$

38)



Oxidation-Half	Reduction-Half
$\# \text{C}_6\text{H}_5\text{NO}_2 \rightarrow \text{CO}_2 + \text{N}_2$ Balance the main atoms $2\text{C}_6\text{H}_5\text{NO}_2 \rightarrow 12\text{CO}_2 + \text{NO}_2 + \text{N}_2$ $\#$ Balance the O-atoms by adding H_2O $\Rightarrow 20\text{H}_2\text{O} + 2\text{C}_6\text{H}_5\text{NO}_2 \rightarrow 12\text{CO}_2 + \text{N}_2$ $\Rightarrow$ Balance the H-atoms by adding H^+ ions. $\Rightarrow 20\text{H}_2\text{O} + 2\text{C}_6\text{H}_5\text{NO}_2 \rightarrow 12\text{CO}_2 + \text{N}_2 + 50\text{H}^+$ $\#$ Balance the charge by adding e^- . $20\text{H}_2\text{O} + 2\text{C}_6\text{H}_5\text{NO}_2 \rightarrow 12\text{CO}_2 + \text{N}_2 + 50\text{H}^+ + 50\text{e}^- \dots(\text{I})$	$\text{O}_2 \rightarrow \text{H}_2\text{O}$ $\text{O}_2 \rightarrow 2\text{H}_2\text{O}$ $\Rightarrow \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ $\Rightarrow 4\text{H}^+ + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ $\Rightarrow 4\text{e}^- + 4\text{H}^+ + \text{O}_2 \rightarrow 2\text{H}_2\text{O} \dots(\text{II})$

$\Rightarrow$ Now add both half reactions; By cancelling the e^-
 (Reaction - (I) $\times$ 2) + (Reaction - (II) $\times$ 25)



(A) In oxidation-Half

Per molecules of $\text{C}_6\text{H}_5\text{NO}_2$: Number of e^- lost = 25

(B) 1 mole $\text{C}_6\text{H}_5\text{NO}_2$ required = $\frac{25}{4} = 6.25$ mole of O_2
 moles of oxygen atoms required = $(6.25 \times 2) = 12.5$ mole
 (B $\rightarrow$ incorrect)

(C) 1 mole $\text{C}_6\text{H}_5\text{NO}_2$ gives $\rightarrow \frac{2}{4}$ moles of N_2

$\square$ Vol. of $\text{N}_2(\text{g}) = \frac{2}{4} \times 22.4 = 11.2$ $\square$ l
 at (1 atm, 273K)

(C $\rightarrow$ incorrect)

(D) $\Rightarrow$ incorrect (H_2O is a liquid not a gas at given condition)

39) Nernst equation :

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{2} \log \frac{[\text{Cd}^{2+}]}{[\text{Cu}^{2+}]}$$

To make E_{cell} less positive, the ratio $\frac{[\text{Cd}^{2+}]}{[\text{Cu}^{2+}]}$ must be increased.

40) A. Incorrect: H_4XeO_6 ($E^{\circ} = 3.0$ V) is a stronger oxidizing agent than F_2 ($E^{\circ} = 2.87$ V).

B. Correct: O_3 ($E^{\circ} = 2.07$ V) can oxidize Cl_2 ($E^{\circ} = 1.36$ V).

C. Incorrect: ClO_4^- has $E^{\circ} = 1.23$ V in acid medium vs 0.36 V in basic medium (better in acidic).

D. Correct: $[\text{Fe}(\text{CN})_6]^{4-}$ ($E^{\circ} = 0.36$ V) is easily oxidized by species with higher reduction potentials like Ce^{4+} (1.67 V) and BrO^- (0.76 V).

41) Cell Reaction: $2\text{Fe}(\text{s}) + \text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\square) \rightarrow 2\text{Fe}^{2+}(\text{aq}) + 4\text{OH}^-(\text{aq})$

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ} = 0.40 - (-0.44) = 0.84 \text{ V}$$

In neutral water, $\text{pH} = 7 \Rightarrow [\text{OH}^-] = 10^{-7} \text{ M}$.

Partial pressure of $O_2 = 0.20$ bar.

Nernst Equation ($n = 4$):

$$E = E^\circ - \frac{0.06}{4} \log \left(\frac{[Fe^{2+}][OH^-]^4}{PO_2} \right)$$

$$E = 0.84 - 0.015 \log \left(\frac{(0.01)^2 (10^{-7})^4}{0.20} \right)$$

$$E = 1.3095 \text{ V}$$

42) 20 mL NaOH added gives $pH = 7.50 = pK_a$, indicating this is the half-equivalence point. Equivalence point occurs at 40 mL NaOH.

At 40 mL, total volume = 40 mL HOCl + 40 mL NaOH = 80 mL.

$$[OCl^-] = \frac{0.2 \times 40}{80} = 0.1M$$

$$pH = 7 + \frac{1}{2}[7.5 + \log 0.1]$$

$$= 10.25$$

43) From graph: $\frac{dE^\circ}{dT} = \frac{0.01}{400} = 2.5 \times 10^{-5} \text{ V/K.}$

At $T = 400 \text{ K}$, $E^\circ = 0.020 \text{ V.}$

Thermodynamic relation ($n = 6$):

$$\Delta H^\circ = -nFE^\circ + nFT \left(\frac{dE^\circ}{dT} \right)$$

$$\Delta H^\circ = -6 \times 96500 \times 0.02 + 6 \times 96500 \times 400 \times 2.5 \times 10^{-5}$$

$$\Delta H^\circ = -6 \times 965 = -5790 \text{ J/mol}$$

44) Reaction: $2X(s) \rightleftharpoons 2Y(g) + Z(g)$

At 300 K: $P_{\text{total}} = 3 \text{ atm} \Rightarrow P_y = 2, P_z = 1 \Rightarrow K_{p1} = 2^2 \times 1 = 4.$

At 600 K: $P_{\text{total}} = 12 \text{ atm} \Rightarrow P_y = 8, P_z = 4 \Rightarrow K_{p2} = 8^2 \times 4 = 256.$

$$\Delta G^\circ = -RT \ln K_p \Rightarrow \frac{\Delta G^\circ}{T} = -R \ln K_p$$

$$\frac{\Delta G_1^\circ}{300} - \frac{\Delta G_2^\circ}{600} = -R \ln(4) - (-R \ln 256) = R \ln \left(\frac{256}{4} \right) = R \ln(64)$$

Thus, $m = 64.$

45) Pure Water: $K_{sp} = 4s^3 = 4 \times 10^{-12} \Rightarrow s = 10^{-4} \text{ M.}$

In $CaCl_2$ solution: New solubility $s' = \frac{10^{-4}}{4 \times 10^8} = 2.5 \times 10^{-13} \text{ M.}$

$$K_{sp} = s' \times [Cl^-]^2 \Rightarrow 4 \times 10^{-12} = (2.5 \times 10^{-13}) \times [Cl^-]^2$$

$$[Cl^-]^2 = 16 \Rightarrow [Cl^-] = 4M$$

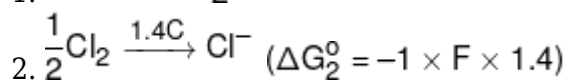
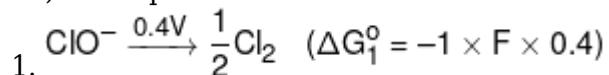
$$\text{Concentration of } CaCl_2 = \frac{[Cl^-]}{2} = \frac{4}{2} = 2 \text{ M}$$

46) Cell constant (G^*): $G^* = \kappa_1 \times R_1 = 1.4 \text{ S cm}^{-1} \times 50 \Omega = 70 \text{ cm}^{-1}$

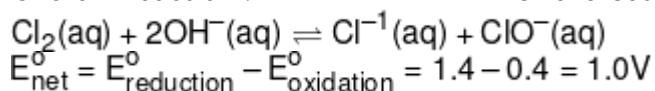
For 0.5 M Solution : $\kappa_2 = \frac{G^*}{R_2} = \frac{70}{280} = 0.25 \text{ S cm}^{-1}$

Molar Conductivity (Λ_m) : $\Lambda_m = \frac{1000 \times \kappa_2}{M} = \frac{1000 \times 0.25}{0.5} = 500 \text{ S cm}^2 \text{ mol}^{-1}$

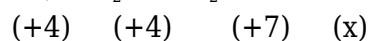
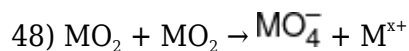
47) Given potentials :



Overall Reaction : $\text{ClO}^- + \text{Cl}^- \rightarrow \text{Cl}_2$ is reversed of disproportionation :

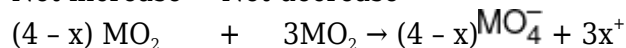


Value of $\left\{ -\frac{10\Delta G^0}{F} \right\} = 10 \times n \times E^0 = 10 \times 1 \times 1.0 = 10\text{V}$



Net increases = 3, Net decrease = (4 - x)

Net increase = Net decrease



↓

↓

(Oxidation part) (Reduction part)

Now, $\frac{4-x}{3} = \frac{2}{3}$

$x = 2$